**Evaluate yourself 3**

3. Using Ellingham diagram (fig 1.4) indicate the lowest temperature at which ZnO can be reduced to Zinc metal by carbon. Write the overall reduction reaction at this temperature.

## 1.5 Electrochemical principle of metallurgy

Similar to thermodynamic principles, electrochemical principles also find applications in metallurgical process. The reduction of oxides of active metals such as sodium, potassium etc., by carbon is thermodynamically not feasible. Such metals are extracted from their ores by using electrochemical methods. In this technique, the metal salts are taken in a fused form or in solution form. The metal ion present can be reduced by treating it with some suitable reducing agent or by electrolysis.

Gibbs free energy change for the electrolysis process is given by the following expression

$$\Delta G^\circ = -nFE^\circ$$

Where n is number of electrons involved in the reduction process, F is the Faraday and $E^0$ is the electrode potential of the redox couple.

If $E^0$ is positive then the $\Delta G$ is negative and the reduction is spontaneous and hence a redox reaction is planned in such a way that the e.m.f of the net redox reaction is positive. When a more reactive metal is added to the solution containing the relatively less reactive metal ions, the more reactive metal will go into the solution. For example,

$$Cu\,(s) + 2Ag^+\,(aq) \longrightarrow Cu^{2+}\,(aq) + 2Ag\,(s)$$

$$Cu^{2+}\,(aq) + Zn\,(s) \longrightarrow Cu\,(s) + Zn^{2+}\,(aq)$$

### 1.5.1 Electrochemial extraction of aluminium - Hall-Heroult process:

In this method, electrolysis is carried out in an iron tank lined with carbon which acts as a cathode. The carbon blocks immersed in the electrolyte act as a anode. A 20% solution of alumina, obtained from the bauxite ore is mixed with molten cryolite and is taken in the electrolysis chamber. About 10% calcium chloride is also added to the solution. Here calcium chloride helps to lower the melting point of the mixture. The fused mixture is maintained at a temperature of above 1270 K. The chemical reactions involved in this process are as follows.

Ionisaiton of alumina $\quad Al_2O_3 \longrightarrow 2Al^{3+} + 3O^{2-}$

Reaction at cathode $\quad 2Al^{3+}\,(melt) + 6e^- \longrightarrow 2Al\,(l)$

Reaction at anode $\quad 6O^{2-}\,(melt) \longrightarrow 3O_2 + 12e^-$

Since carbon acts as anode the following reaction also takes place on it.



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